SCHOOL OF COMPUTING, TECHNOLOGY AND APPLIED SCIENCES
COMPUTER SCIENCE DEPARTMENT

END OF SEMESTER EXAMINATIONS

COURSE: DATABASE SYSTEMS
COURSE CODE: CIT 2641

TIME ALLOWED: 3 Hours
TOTAL MARKS: 100 Marks

GENERAL INSTRUCTIONS TO CANDIDATES:

1. This paper contains **FIVE (5)** questions, each carrying **25 marks.**
2. Answer **ANY FOUR (4)** questions.
3. Each question has sub-questions labelled (a) to (e), each carrying **5 marks.**
4. **Neat and clear work** is highly recommended.
5. Write your student number clearly on all answer sheets.
6. Begin each question on a new page.
7. Where SQL code or diagrams are required, ensure they are clearly labelled and legible.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

QUESTION 1

(a) Explain why software is NOT classified as data, and discuss the statement: "data is the technological term and information is the business term." ***[5 marks]***

(b) Traditional file-based processing systems suffered from several well-documented problems. Describe FIVE of these problems in detail, explaining for each problem: (i) what the problem is, (ii) how it arises in a file-based environment, and (iii) what impact it has on an organisation's operations. Use a realistic business scenario to illustrate your answer. ***[5 marks]***

(c) Explain what a Database Management System (DBMS) is, and describe how it addresses the shortcomings of traditional file-based systems. In your answer, define the terms Database (DB), meta-data, Data Dictionary (DD), and database table. ***[5 marks]***

(d) Database users are classified into two broad groups: Actors on the Scene and Workers Behind the Scene. For EACH group, identify at least THREE specific types of users, describe their roles and responsibilities within the database environment, and explain how their work differs from one another. Pay particular attention to the distinction between the Data Administrator (DA) and the Database Administrator (DBA). ***[5 marks]***

(e) A university is currently using separate spreadsheet files maintained by its Admissions, Finance, and Academic departments. The Admissions department stores student contact details, Finance stores fees payment records, and Academic Records stores course enrolments. Identify FOUR specific problems this arrangement causes, and then propose and justify a database solution that would resolve these problems. Your answer should reference specific database concepts such as data redundancy, data consistency, and data sharing. ***[5 marks]***

QUESTION 2

(a) The ANSI-SPARC three-level schema architecture is a foundational concept in database design. Describe each of the three levels External, Conceptual, and Internal in detail. For each level, explain: (i) what it describes, (ii) who uses it and why, (iii) the schema associated with it, and (iv) an example of what its description might look like for a student enrolment system. ***[5 marks]***

(b) Data independence is one of the key advantages of the database approach. Distinguish carefully between logical data independence and physical data independence. ***[5 marks]***

(c) Data models are used to describe and structure data. Compare and contrast the following three categories of data models: (i) Conceptual data models (ii) Representational/Implementation data models (e.g., the Relational model), and (iii) Physical data models. In your comparison, address the purpose of each, the level of abstraction, the intended audience, and the stage of database development at which each is used. ***[5 marks]***

(d) Explain the following four key characteristics of the database approach, providing a practical example to illustrate each one: (i) the self-describing nature of a database system and the role of meta-data, (ii) program-data independence and data abstraction, (iii) support for multiple views of data, and (iv) data sharing and multi-user transaction processing. For characteristic (iv), also explain why concurrency control is necessary. ***[5 marks]***

(e) A company's employee database was designed three years ago. Since then, a new attribute 'NationalIDNumber' has been added to the Employee table, and the 'Salary' attribute has been moved to a new confidential table. (i) How does the schema change in each case? (ii) Which type of data independence logical or physical is being exercised in each case? (iii) What must remain unchanged at the external level for data independence to hold? ***[5 marks]***

QUESTION 3

(a) Explain the fundamental concepts of the Entity-Relationship (ER) model by defining and giving examples of each of the following: (i) entity type and entity instance, (ii) attribute types simple, composite, multi-valued, and derived attributes, (iii) key attribute and its purpose, (iv) relationship type and relationship instance, and (v) weak entity type and its identifying relationship. Use a university database scenario (students, courses, lecturers) to illustrate each concept. ***[5 marks]***

(b) Cardinality and participation constraints are critical in ER modelling. Define and distinguish between: (i) one-to-one (1:1), one-to-many (1:N), and many-to-many (M:N) relationship cardinalities, (ii) total (mandatory) participation and partial (optional) participation. For each of the following relationships in a hospital database, state the cardinality and participation constraints and justify your answer: (A) a Doctor TREATS a Patient, (B) a Ward CONTAINS Beds, (C) a Patient is ASSIGNED-TO a Ward, (D) a Nurse SUPERVISES other Nurses. ***[5 marks]***

(c) Compare and contrast the top-down and bottom-up approaches to data modeling. Describe the step-by-step process for each approach. ***[5 marks]***

(d) A retail company wants to build a database to manage its operations. The following information has been gathered from interviews with staff:

- The company sells products, each identified by a product code, name, and unit price.
- Products are supplied by suppliers, each with a unique supplier ID and contact information. A product can be supplied by multiple suppliers, and each supplier can supply many products.

- Customers place orders. Each order has an order ID, order date, and can contain multiple products in specified quantities.

- Each order is handled by exactly one sales employee. (i) Identify all entity types, their attributes, and their key attributes. (ii) Identify all relationships, their cardinalities, and participation constraints. (iii) Draw a complete ER diagram for this scenario. ***[10 marks]***

QUESTION 4

(a) The relational model is based on a formal mathematical foundation. Define each of the following relational concepts and explain its significance: (i) relation, tuple, and attribute, (ii) domain and degree of a relation, (iii) primary key, candidate key, and foreign key, (iv) referential integrity constraint and entity integrity constraint, (v) the difference between a relation schema and a relation instance. ***[5 marks]***

(b) Use a simple EMPLOYEE(EmpID, Name, DeptID, Salary) and DEPARTMENT(DeptID, DeptName, ManagerID) schema to illustrate each concept in (a). ***[5 marks]***

(c) SQL includes Data Definition Language (DDL) and Data Manipulation Language (DML) statements. Using the schema below, write SQL statements to accomplish each of the following tasks: STUDENT(StudentID, FirstName, LastName, DateOfBirth, ProgramID, GPA) PROGRAM(ProgramID, ProgramName, FacultyID)

(i) Create the STUDENT table with appropriate data types, a primary key, a foreign key to PROGRAM, and a CHECK constraint ensuring GPA is between 0.0 and 4.0. (ii) Insert three student records into the STUDENT table.

(iii) Retrieve the full names and GPAs of all students enrolled in 'Computer Science' whose GPA exceeds 3.5, ordered by GPA descending.

(iv) Update the GPA of student with StudentID = 'S001' to 3.8. (v) Delete all students who have no program assigned (ProgramID IS NULL).

[10 marks]

(c) Advanced SQL queries often require joins, subqueries, and aggregate functions. Using the following schema, write complete SQL queries for each task: EMPLOYEE(EmpID, EmpName, Salary, DeptID, ManagerID) DEPARTMENT(DeptID, DeptName, Location) PROJECT(ProjID, ProjName, DeptID) WORKS_ON(EmpID, ProjID, Hours) (i) List the names of all employees who work on MORE THAN two projects.

(ii) Find the name and average salary of each department, but only display departments where the average salary exceeds 50,000.

(iii) List employees who earn more than the average salary of their own department (use a correlated subquery). ***[5 marks]***

QUESTION 5

(a) Functional dependencies (FDs) are the foundation of normalisation theory. Define the concept of a functional dependency (FD) and explain what it means for one attribute to functionally determine another. ***[5 marks]***

(b) Normalisation is a systematic process of organising data to reduce redundancy and improve integrity. Describe the process of normalising the following relation step-by-

step from an unnormalised form (UNF) through First Normal Form (1NF), Second Normal Form (2NF), and Third Normal Form (3NF). ***[5 marks]***

(c) Boyce-Codd Normal Form (BCNF) is a stricter version of 3NF. Explain the difference between 3NF and BCNF, and describe the specific situation in which a relation can be in 3NF but NOT in BCNF. ***[5 marks]***

(d) PL/SQL is Oracle's procedural extension to SQL. With the use of an appropriate example, explain the purpose of PL/SQL and how it extends the capabilities of standard SQL. ***[5 marks]***

END OF EXAMINATION PAPER

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